DRIVER DROWSINESS DETECTION

(B. Tech. Project-2)
A REPORT
Submitted by

Lavi Garg

(21ITUON138)

Charmi Parmar

(21ITUBS073)

for the partial fulfilment of the requirements for Semester –VII of

BACHELOR OF TECHNOLOGY

(INFORMATION TECHNOLOGY)

Under the guidance of

Prof. Archana N. Vyas

Dharmsinh Desai University, Nadiad.

Department of Information Technology

Faculty of Technology,

DHARMSINH DESAI UNIVERSITY

NADIAD 387001

October, 2024

CANDIDATE DISCLOSURE ON THE USE OF AI TOOLS

In the process of writing this report, we used the following AI tools and technologies:

- ChatGPT (by OpenAI) was used to assist in generating the structure and content of this report, including sections like the abstract, introduction, literature review, and analysis of findings.
- Gemini (Google's AI tool) was utilized for conducting research, gathering relevant information, and identifying key studies related to driver drowsiness detection.
- Grammarly (Free version) was used to correct errors in spelling, grammar, and mechanics.
- Google Colab was used to implement and execute the code for our CNN model for driver drowsiness detection.

CANDIDATE'S DECLARATION

We declare that the dissertation (for B.Tech in Information Technology) titled " Driver Drowsiness Detection " is our own work being conducted under the guidance and supervision of Prof. Archana N. Vyas.

We further declare that to the best of our knowledge, this dissertation does not contain any part of work which has been submitted for the award of any degree either in this University or in any other University without proper citation.

Lavi Garg

Charmi Parmar

CERTIFICATE

This is to certify that this Report of B. Tech. Project2 submitted for partial fulfilment of B. Tech Semester- VII is a record of the work carried out by

1) LAVI GARG

   ID No. 21ITUON138,

   B. Tech. Sem – VII (Information Technology): 2024-25

2) CHARMI PARMAR

   ID No. 21ITUBS073,

   B. Tech. Sem – VII (Information Technology): 2024-25

Guide
Prof. Archana N. Vyas
Assistant Professor,
Department of Information Technology
Dharmsinh Desai University,
Nadiad–387001, INDIA

HoD
Prof. Dr. V. K. Dabhi
Head, Dept. of Information Technology
Faculty of Technology
Dharmsinh Desai University
Nadiad–387001, INDIA

Department of Information Technology

Faculty of Technology

Dharmsinh Desai University

College Road, Nadiad-387001, INDIA

ACKNOWLEDGMENT

We extend our deepest gratitude to the IT Department for providing us with the opportunity and resources to undertake this research. Their unwavering support and guidance have been invaluable throughout the journey.

We owe special thanks to Prof. Archana N. Vyas, whose mentorship has been instrumental in the development of our Driver Drowsiness Detection research. Her expert knowledge, encouragement, and insightful feedback have significantly influenced our work and professional growth, motivating us to push beyond our limits.

We also wish to thank everyone who generously contributed their images for the creation of our dataset. Without their willingness and support, this project would not have been possible.

Lastly, we are profoundly grateful to our families, especially our parents, for their enduring support and encouragement. Their belief in us has been a constant source of strength.

Thank you to all who have supported and guided us in this journey—your contributions have made this research a truly rewarding experience.

Lavi Garg
Dharmsinh Desai University, Nadiad
October, 2024
21ituon138@ddu.ac.in

Charmi Parmar
Dharmsinh Desai University, Nadiad
October, 2024
21itubs073@ddu.ac.in

ABSTRACT

DRIVER DROWSINESS DETECTION

Project2 by Lavi Garg and Charmi Parmar

at

Dharmsinh Desai University, October 2024

Driver drowsiness is a critical factor in traffic accidents globally, emphasizing the need for reliable detection mechanisms to enhance road safety. This research introduces a Driver Drowsiness Detection model based on a custom Convolutional Neural Network (CNN) architecture, designed to identify drowsiness from static facial images. To address limitations in existing datasets, a unique dataset was created with 72.59% self-collected data from 27 subjects (14 males, 13 females), including 300 images of bus drivers to represent diverse real-world conditions. The remaining 27.41% of the dataset was sourced from publicly available databases, ensuring greater diversity in the data.

The CNN model incorporates convolutional layers with ReLU activation, max pooling, fully connected layers, dropout for overfitting prevention, and the Adam optimizer with categorical cross-entropy as the loss function. Experimental results demonstrate that the custom model outperforms pre-trained models (ResNet, VGG, InceptionV3), achieving an accuracy of 81.73% and an ROC AUC score of 0.82, underscoring its ability to detect drowsy states accurately.

While promising, the model currently operates only on static images, limiting its applicability in real-time scenarios. Future work will focus on extending the model to process video streams, allowing for real-time drowsiness detection and continuous driver monitoring. This project lays a strong foundation for advanced drowsiness detection systems, with the potential to significantly reduce accident risks and improve on-road safety.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION

1.1 Introduction to the research problem

Driver drowsiness is a significant safety concern on the road, often leading to accidents. Traditional methods of detecting driver fatigue, such as manual observation or physiological sensors, are often unreliable or invasive. This research aims to address these limitations by leveraging the power of deep learning and computer vision to develop a non-invasive and accurate system for image-based driver drowsiness detection.

1.2 Motivation for research work

The motivation behind this research stems from the alarming statistics surrounding driver fatigue-related accidents. By developing a reliable and efficient image-based driver drowsiness detection system, we can potentially:

- **Provide a foundation for future research:** Serve as a basis for developing real-time video-based systems.
- **Enhance understanding:** Gain insights into the facial cues associated with driver fatigue.
- **Inform the development of assistive technologies:** Contribute to the creation of systems that can alert drivers when they exhibit signs of drowsiness.

1.3 Objectives and scope of the research work

The primary objective of this research is to develop a deep learning-based computer vision model that can accurately detect signs of driver drowsiness from static images. The scope of this research includes:

- **Data collection:** Acquiring a diverse dataset of driver images that captures various facial expressions, lighting conditions, and facial accessories/headwear.
- **Model development:** Designing and training a deep learning model capable of extracting relevant features from the images and classifying them as drowsy or awake.
- **Model evaluation:** Assessing the performance of the developed model using appropriate metrics such as accuracy, precision, recall, and F1-score.
- **Future directions:** Discussing the potential challenges and opportunities for extending the model to real-time video-based applications.

2. BACKGROUND THEORY

2.1 Deep Learning in Computer Vision

Deep learning has revolutionized the field of artificial intelligence by enabling models to learn automatically from large datasets. At the core of deep learning are neural networks, which consist of layers of interconnected neurons designed to mimic the human brain's processing. These networks are particularly effective in tasks involving large-scale data, such as image recognition, natural language processing, and autonomous systems.

In the case of driver drowsiness detection, deep learning techniques analyze visual inputs- typically images of a driver's face- to detect signs of fatigue. Unlike traditional rule-based approaches, which rely on manually extracted features, deep learning models, particularly Convolutional Neural Networks (CNNs), can learn relevant features from raw data. These features might include indicators like eye closure, head tilting, and yawning, which signal drowsiness. This ability to recognize subtle facial expressions makes deep learning well-suited for driver monitoring systems.

The scalability and adaptability of deep learning models are crucial in real-world applications, where conditions can vary widely. For driver drowsiness detection, models need to handle different lighting conditions, facial obstructions, and varied camera angles. With training on a comprehensive dataset, deep learning models can offer real-time solutions that outperform traditional methods.

2.2 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are specialized deep learning models designed for grid-like data, such as images. CNNs have proven highly effective in image classification, object detection, and face recognition tasks due to their hierarchical structure, which extracts features at various levels of abstraction.

A typical CNN is composed of the following layers:

- **Convolutional Layers**: These layers apply filters to detect features such as edges, corners, textures, and more complex patterns as the network deepens. This enables the model to learn abstract features, such as facial expressions indicative of drowsiness.
- **Pooling Layers**: Pooling reduces the dimensionality of feature maps, focusing on the most important information while lowering computational costs. Max-pooling is commonly used, selecting the most prominent feature from a given region of the image.
- **Activation Functions**: Non-linear activation functions, like ReLU (Rectified Linear Unit), introduce non-linearity, enabling the network to learn complex

patterns. Without activation functions, the network would behave like a linear model, limiting its ability to capture intricate data relationships.

- **Fully Connected Layers**: After the convolutional and pooling layers, fully connected layers interpret the extracted features and make a final classification. These layers map the learned features to output classes like "drowsy" or "awake."

In this project, CNNs were applied to static images of drivers to classify them as drowsy or awake. The CNN architecture's ability to capture both local and global patterns within images made it an ideal choice for analyzing facial features that indicate drowsiness.

Although CNNs excel at image classification, they require large datasets for optimal performance. This led to the exploration of transfer learning using pre-trained models.

2.3 Transfer Learning and Pre-trained Models

Transfer Learning is a technique where a model trained for one task is repurposed for another, often related task. In deep learning, this typically involves using a pre-trained model that has learned from a large dataset (e.g., ImageNet) and fine-tuning it for a specific task.

For this project, pre-trained models such as ResNet50, InceptionV3, and VGG16 were initially used as starting points. These models are well-known for their success in large-scale image classification tasks:

- **ResNet50 (Residual Networks)**: ResNet50 introduces residual learning, enabling very deep networks to train effectively without suffering from the vanishing gradient problem. Skip connections allow layers to bypass certain stages, maintaining high accuracy in deep architectures.
- **InceptionV3**: InceptionV3 is designed to balance network depth and computational efficiency by using a variety of filter sizes at each layer. This multi-scale processing enables the model to capture different types of features, from small details to larger patterns, within the same layer.
- **VGG16**: VGG16 is characterized by its simplicity, using small 3x3 convolution filters in a deep architecture. Despite its straightforward design, VGG16 achieves impressive results in image classification tasks, making it a strong candidate for transfer learning applications.

These models were fine-tuned using the custom driver drowsiness detection dataset. However, they did not perform as well as expected, likely due to the specific challenges posed by the dataset, such as facial obstructions, low lighting, and varied camera angles. As a result, a custom-built CNN model was developed to better handle the dataset's unique characteristics.

2.4 Dataset Creation for Driver Drowsiness Detection

The creation of a custom dataset was a crucial step in the success of this project. Many existing datasets for driver drowsiness detection suffer from several limitations:

- **Limited Variability**: Publicly available datasets often fail to account for real-world variability, such as different lighting conditions (day vs. night) or facial obstructions (e.g., sunglasses, masks).
- **Inconsistent Angles**: Most datasets assume a fixed camera position directly in front of the driver, which is not always the case in real-world monitoring systems.
- **Class Imbalance**: Many datasets are biased toward "awake" images, as drivers typically spend more time awake than drowsy.

To address these issues, a custom dataset was created to capture drivers under various conditions:

- **Lighting Conditions**: Images were taken in both well-lit and dim environments, simulating driving conditions throughout the day and night.
- **Facial Accessories**: Drivers wore sunglasses, hats, and masks to simulate real-world obstructions that might interfere with facial recognition.
- **Multiple Camera Angles**: The dataset included images captured from various angles, ensuring that the model could handle different camera setups in real vehicles.

This dataset was vital for training the custom CNN, as it provided the diversity needed to ensure robust generalization to different environments.

2.5 Custom CNN Model and Data Augmentation

After testing pre-trained models and recognizing their limitations, a custom CNN model was developed to address the unique challenges of the dataset. The custom CNN incorporated multiple layers to capture the diverse patterns and features present in the images.

The architecture included convolutional layers to extract features, pooling layers to reduce dimensionality, and fully connected layers for classification. The model was designed to classify drivers as “drowsy” or “awake.”

Data Augmentation techniques were used extensively to enhance the model’s robustness. This included converting images from RGB to grayscale, adjusting brightness, and applying random rotations to make the model adaptable to different lighting conditions and orientations. These techniques helped the custom CNN perform better than the pre-trained models on both the training and test datasets.

2.6 Importance of Data Diversity

A critical aspect of developing a reliable drowsiness detection system is ensuring the dataset's diversity to cover a wide range of real-world conditions. Without sufficient diversity, models risk overfitting to specific conditions, limiting their generalization to new situations.

The custom dataset addressed this issue by including varied lighting, facial obstructions, and different camera angles. This diversity was crucial for the custom CNN model to achieve high accuracy and robust performance in real-world applications.

3. LITERATURE REVIEW

3.1 Overview of Driver Drowsiness Detection Research

Research on driver drowsiness detection has evolved significantly over the years. Early efforts primarily focused on image processing techniques that monitored eye blinking, head movements, and yawning to detect signs oratf fatigue. These systems aimed to provide real-time alerts to prevent accidents caused by drowsy driving. However, the detection accuracy of these approaches was often limited by environmental factors, such as lighting conditions and camera angles.

As technology advanced, the field shifted towards machine learning methods that improved detection accuracy by allowing systems to learn from data. The rise of smartphone technology and in-vehicle cameras enabled the development of portable systems capable of continuous monitoring, often incorporating biometric measures for enhanced reliability.

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), has further transformed the landscape by enabling the automatic learning of complex features from large datasets. These models can analyze facial expressions and eye states in real-time, pushing the boundaries of detection capabilities. However, challenges remain in balancing accuracy with real-time performance, leading to ongoing research focused on refining algorithms, expanding datasets, and integrating multiple sensing modalities for comprehensive driver monitoring.

3.2 Traditional Approaches to Drowsiness Detection

Previous studies employed methods like eye-tracking and facial landmark detection to monitor drivers' fatigue. These approaches relied on manually designed features and rule-based systems, which were heavily dependent on consistent external conditions, such as lighting. While useful, these methods faced challenges in real-world applications, particularly under low-light conditions or when drivers' facial features were partially obscured.

3.3 Machine Learning Approaches for Driver Drowsiness Detection

Recent advancements in driver drowsiness detection increasingly utilize machine learning techniques to improve the accuracy and efficiency of these systems. Various studies have demonstrated the effectiveness of different algorithms and methodologies in detecting signs of drowsiness in real time.

Ayman Altameem et al.[1] developed a driver drowsiness detection system using Support Vector Machines (SVM) based on facial expressions. Tested under varying luminance, it achieved 83.25% accuracy in detecting facial expression changes. The system was evaluated on a custom driver image dataset, proving effective for real-time drowsiness detection and road safety enhancement.

Yaman Albadawi et al.[2] reviewed different driver drowsiness detection systems, including image-based, biological-based, vehicle-based, and hybrid measures. They highlighted Knapik and Cyganek's thermal imaging system (87% accuracy) and Kiashari et al.'s facial thermal analysis (90% accuracy with SVM). The review underscored the effectiveness of integrating multiple features for better real-time detection.

Yaman Albadawi et al.[3] also created a real-time driver drowsiness detection system using Eye Aspect Ratio (EAR) and head pose estimation. Tested on the National Tsing Hua University Driver Drowsiness Detection dataset, it achieved 99% accuracy using classifiers like random forest and SVM, showcasing the effectiveness of image-based methods for non-invasive driver monitoring.

3.4 CNN Models for Driver Drowsiness Detection

Recent advances in driver drowsiness detection systems have utilized both pretrained CNN models, such as ResNet50, VGGNet, and InceptionV3, and custom-built CNN architectures. Pretrained models can be fine-tuned using transfer learning for drowsiness detection tasks, allowing high accuracy with less training data. Additionally, researchers have designed their own CNN models to optimize detection by focusing on specific facial features like eye closure and yawning, achieving reliable real-time performance. This combination of transfer learning and custom CNN models has enhanced the effectiveness of drowsiness detection systems.

Hashemi et al.[4] introduced a CNN-based system focused on eye closure detection for drowsiness. Their FD-NN model, tested on the ZJU dataset, achieved 98.15% accuracy with low computational complexity, making it effective for real-time applications.

Rateb Jabbar et al.[5] developed a driver drowsiness detection model using Convolutional Neural Networks (CNNs) with facial landmark detection, achieving 83.3% accuracy. Utilizing the NTHU Driver Drowsiness Detection Dataset, this lightweight model is designed for mobile and embedded systems, offering a promising real-time solution for enhancing road safety.

Salman et al.[6] created an ensemble CNN (ECNN) model focused on facial features like yawning and eye closure. Tested on the YawDD dataset, the ensemble outperformed individual CNNs, achieving up to 99.42% accuracy, demonstrating the effectiveness of ensemble techniques.

Md. Tanvir Ahammed Dipu et al.[7] proposed a driver drowsiness detection system using MobileNet with Single Shot Multibox Detector (SSD). Trained on a dataset of 4,500 labeled images, the system achieved a mean average precision (mAP) of 98.02% in detecting yawning and eye states, enhancing road safety by alerting drowsy drivers.

Minhas et al.[8] developed a CNN-based driver drowsiness detection system, testing InceptionV3, VGG16, and ResNet50 on a custom dataset. ResNet50 achieved the highest accuracy of 93.69%, showing robust real-time performance under various lighting conditions, making it suitable for alerting drowsy drivers.

Jahan et al.[9] created a driver drowsiness detection system utilizing a custom 4D CNN model, alongside VGG16 and VGG19, tested on the MRL Eye dataset. The 4D model achieved the highest accuracy of 97.53%, while the VGG models reached 95.93% and 95.03%, respectively. This system enables real-time drowsiness detection using just a webcam, providing alerts to drivers.

Esra Kavalcı Yılmaz and M. Ali Akcayol[10] introduced the SUST-DDD dataset for driver drowsiness detection, achieving 90.53% accuracy with a VGG19+LSTM model. The dataset includes 2,074 videos of drivers in drowsy and alert states, highlighting the potential of hybrid models to improve driver safety.

Ajinkya Rajkar et al.[11] created a driver drowsiness detection system using CNNs to analyze facial features, focusing on eye states and yawning. They achieved an impressive average accuracy of 96% with the Closed Eye in the Wild (CEW) and Yawning Detection Dataset (YawDD), effectively processing live video streams for real-time alerts.

Florez et al.[12] developed a CNN-based system using InceptionV3, VGG16, and ResNet50V2 on the NITYMED dataset. ResNet50V2 achieved 99.71% accuracy, using MediaPipe for eye detection and transfer learning for real-time drowsiness detection.

Nandyal and Sharanabasappa[13] developed a driver drowsiness detection system using the ResNet 18 model, enhanced by firefly optimization techniques. Achieving 91.3% accuracy on the NTHU-DDD dataset, the system effectively addresses challenges like vanishing gradients and overfitting. Future improvements aim to incorporate transfer learning for more accurate drowsiness detection.

Mohammed Imran Basheer Ahmed et al.[14] developed a driver drowsiness detection system using CNNs and VGG16, analyzing facial expressions and eyeball movements. Tested on 2,900 images, the CNN model achieved 97% accuracy, demonstrating real-time drowsiness detection potential.

Jebraeily et al.[15] used a genetic algorithm to optimize CNN structures for drowsiness detection, achieving 99.8% accuracy. The system was trained on a combined dataset and improved through transfer learning, showing potential for real-time applications.

Salem and Waleed[16] developed a deep learning-based system using MobileNetV2 and InceptionV3. Their model achieved impressive accuracy rates of 99.94% and 99.82% on the Driver Drowsiness Dataset, making it highly effective for real-time detection.

Literature Review Summary

Title Paper	Author(s)	Method	Key Features	Dataset	Accuracy
Early Identification and Detection of Driver Drowsiness by Hybrid Machine Learning [1]	A. Altameem, S. Kumar, A. Rameshchandra poonia, A. Jilani Saudagar	SVM	Detects drowsiness through facial expressions	Custom Driver Image Dataset	83.25%
A Review of Recent Developments in Driver Drowsiness Detection Systems [2]	Y. Albadawi, M. Takruri, M. Awad	SVM	Discusses various techniques for driver drowsiness detection	Multiple datasets	90%
Real-Time Machine Learning-Based Driver Drowsiness Detection Using Visual Features [3]	Y. Albadawi, A. AlRedhaei, M. Takruri	Eye Aspect Ratio (EAR), Head Pose Estimation, SVM, Random Forest	Real-time drowsiness detection using facial features	NTHU Driver Drowsiness Dataset	99%
Driver Safety Development: Real-Time Driver Drowsiness Detection System Based on Convolutional Neural Network [4]	M. Hashemi, A. Beheshti Shirazi	CNN (FD-NN model)	Focus on eye closure for drowsiness detection	ZJU Dataset	98.15%
Driver Drowsiness Detection Model Using Convolutional Neural Networks Techniques for Android Application [5]	R. Jabbar, M. Shinoy, M. Kharbeche, K. Al-Khalifa, M. Krichen, K. Barkaoui	CNN for Facial Landmark Detection	Lightweight CNN model for mobile and embedded systems	NTHU Driver Drowsiness Dataset	83.3%
Driver Drowsiness Detection Using Ensemble Convolutional	R. M. Salman, M. Rashid, R. Roy, M. Ahsan, Z. Siddique	Ensemble CNN	Uses facial features like yawning and eye	YawDD Dataset	99.42%

Neural Networks On YAWDD [6]			closure for detection		
Real-time Driver Drowsiness Detection using Deep Learning [7]	Md. T. A. Dipu, S. S. Hossain, Y. Arafat, F. B. Rafiq	Deep Learning, SSD with MobileNet	Real-time drowsiness detection using eye and yawn states	4,500 labeled images dataset	98.02%
A smart analysis of driver fatigue and drowsiness detection using convolutional neural networks [8]	A. A. Minhas, S. Jabbar, M. Farhan, M. N. ul Islam	CNN (Inception V3, VGG16, ResNet50)	Analysis of driver fatigue with CNNs	Custom dataset	93.69%
4D: A Real-Time Driver Drowsiness Detector Using Deep Learning [9]	I. Jahan, K. M. A. Uddin, S. A. Murad, M. Masud, S. Aljahdali, M. S. U. Miah, A. K. Bairagi	4D CNN Model, VGG16, VGG19	Real-time detection using webcam	MRL Eye Dataset	97.53%
SUST-DDD: A Real-Drive Dataset for Driver Drowsiness Detection [10]	E. Kavalcı Yılmaz, M. A. Akcayol	VGG19+LSTM	Hybrid model using LSTM for enhanced accuracy	SUST-DDD Dataset (2,074 videos)	90.53%
Driver Drowsiness Detection Using Deep Learning [11]	A. Rajkar, N. Kulkarni, A. Raut	CNNs for Eye States and Yawning	Focus on eye states and yawning for real-time alerts	CEW and YawDD Dataset	96%
A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification [12]	R. Florez, F. Palomino-Quispe, T. Paixão, A. Alvarez, R. J. Coaquira-Castillo, J. C. Herrera-Levano	CNN (Inception V3, VGG16, ResNet50 V2)	Uses eye state identification for drowsiness detection	NITYMED Dataset	99.71%
Deep ResNet 18 and Enhanced Firefly Optimization Algorithm for On-Road Vehicle Driver Drowsiness Detection [13]	S. Nandyal, S. Sharanabasappa	ResNet18, Firefly Optimization	Focus on optimization techniques for better accuracy	NTHU-DDD Dataset	91.3%

A Deep-Learning Approach to Driver Drowsiness Detection [14]	M. I. B. Ahmed, H. Alabdulkarem, F. Alomair, M. Alahmari, M. Alhumaidan, S. Alrassan, A. Rahman, D. Aldossary, M. Youldash, G. Zaman	CNN, VGG16	Detects facial movements for real-time alerts	2,900 images dataset	97%
Driver Drowsiness Detection Based on Convolutional Neural Network Architecture Optimization Using Genetic Algorithm [15]	Y. Jebraeily, Y. Sharafi, M. Teshnehlab	CNN	Optimized CNN for better accuracy	FER-2013 dataset and a newly created drowsiness dataset	99.8%
Drowsiness Detection in Real-Time via Convolutional Neural Networks and Transfer Learning [16]	D. Salem, M. Waleed	CNN with Transfer Learning	Transfer learning used for real-time drowsiness detection	Driver Drowsiness Dataset	99.94%

3.5 Real-Time Drowsiness Detection and Challenges

Implementing drowsiness detection systems in real-time poses critical challenges for researchers. CNN-based models are often computationally demanding, complicating real-time processing with limited onboard hardware in vehicles. Furthermore, ensuring robust performance across varying conditions, such as changes in lighting or camera angles, remains a challenge. Ongoing efforts focus on optimizing CNN models for faster processing and reducing resource requirements for real-time deployment.

3.6 Limitations of Existing Research

Existing research on driver drowsiness detection has several limitations. Many studies use small datasets that don't represent different driving situations or driver types, which makes it hard to apply their findings in real life. Most research focuses only on visual data, ignoring other important signs like physical reactions or vehicle behaviour. Additionally, the complex algorithms used, especially CNNs, require a lot of computing power, making it difficult to use them in regular cars. There are also concerns about driver privacy when monitoring them continuously. Finally, without standard testing methods, it’s tough to compare different systems effectively. Addressing these issues is crucial for improving drowsiness detection systems.

3.7 Conclusion

This literature review highlights the evolution of driver drowsiness detection techniques, transitioning from early image processing methods to advanced deep learning models like CNNs. Recent advancements have shown that CNNs are highly effective in detecting drowsiness, achieving significantly higher accuracy than traditional methods. Their ability to learn complex features from images makes them the best choice for real-time applications in diverse driving conditions. Despite the challenges in real-time implementation and the need for more comprehensive datasets, the continued development of CNN-based systems holds great promise for enhancing road safety and improving driver monitoring.

4. ANALYSIS AND FINDINGS

In this section, we analyze the performance of the various models and methods applied to our custom driver drowsiness detection dataset. The models evaluated include pre-trained networks (ResNet50, InceptionV3, VGG16) and a custom-built Convolutional Neural Network (CNN).

4.1 Dataset Details

The dataset utilized in this study was developed by combining a custom-created dataset with data from other publicly available sources, namely Driver Drowsiness Dataset (DDD) and Driver Drowsiness Detection -Dataset both from Kaggle. This approach significantly increased the diversity and real-world applicability of the data, accounting for:

- Varied lighting conditions- Both well-lit and low-light images (day and night).
- Facial obstructions- Sun goggles, caps, face masks, and spectacles.
- Camera Angles: Photos from different camera angles to simulate real-world conditions.
- Class Balance: Ensuring a balanced representation of drowsy and awake drivers to prevent bias.

This merging of data sources allowed for more robust model training, particularly for conditions often underrepresented in standalone datasets.

There are two classes in the dataset, “Awake” and “Drowsy”. Each class has 5000 images. Approximately 7,258 images of the dataset were self-collected, while the remaining 2,742 images were drawn from the external datasets.

The self-collected images included data from 27 subjects (14 males, 13 females), with ages ranging from 20 to 60. All subjects were seated in a car during the data capture, with around 300 images specifically featuring a bus driver seated in a bus. The external data helped to address class imbalance and contributed to a more varied representation of drowsy and awake states, ensuring that the model generalizes well across different real-world conditions.

Here are some sample images:

Figure 1: Awake

Figure 2: Drowsy

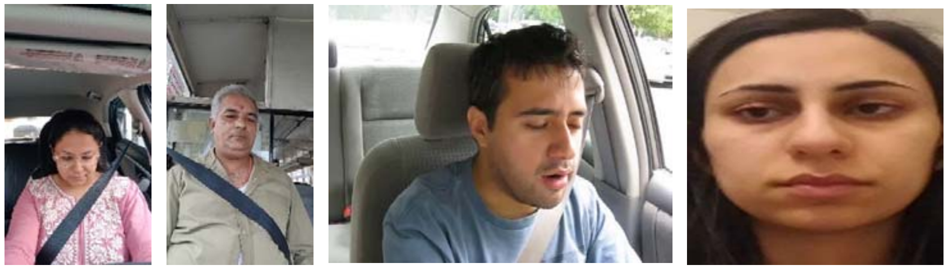


Image Preprocessing:

One of the challenges encountered during data collection and preprocessing was the large image size, which initially slowed down the processing time.

To address this issue, image sizes were reduced, ensuring that all images were kept below 500 MB. Dimensionality was also reduced in some cases, which helped streamline data processing and speed up training. This step was critical to making the model training more efficient without sacrificing accuracy.

4.2 Model Analysis

In this study, four models were chosen to evaluate their performance on the task of driver drowsiness detection: **ResNet50**, **InceptionV3**, **VGG16**, and a **custom CNN model**. These models were selected for their unique architectures, which are well-suited to capturing visual patterns relevant to facial states and expressions in various conditions.

Model-Specific Analysis:

ResNet50: ResNet50 is a deep convolutional neural network with residual connections that allow it to mitigate issues related to vanishing gradients, making it highly effective in deep architectures. Its residual blocks make it easier to train, even at significant depth, enabling it to capture intricate visual details. This model is particularly advantageous when aiming for robust performance across diverse datasets, as the residual connections contribute to better feature propagation and reduced risk of overfitting. However, ResNet50's computational intensity can lead to longer training times and increased resource requirements, which may be challenging for some hardware configurations.

InceptionV3: The InceptionV3 model was chosen for its efficient multi-scale architecture, which processes different receptive fields simultaneously, making it highly effective for detecting features across varying scales. This architecture provides strong performance with fewer parameters than similarly deep networks, making it comparatively lighter and faster to train. In practical terms, InceptionV3's design allows it to handle facial obstructions, such as sunglasses or face masks, more flexibly.

However, its complex module structure can be more challenging to fine-tune for specific characteristics of this dataset, potentially requiring additional adjustments for optimal results.

VGG16: Known for its simpler and highly consistent architecture, VGG16 is well-regarded for its strong feature extraction capabilities, which excel in detecting fine-grained visual details, making it well-suited for tasks that involve facial expressions. VGG16 was chosen because of its uniform 3x3 convolution filters, which enhance the model's ability to generalize effectively. The primary limitation of VGG16, however, is its relatively high memory and computational requirements due to a large number of parameters, making it slower and more resource-intensive to train compared to some other architectures.

Custom CNN Model: The custom CNN model was developed specifically to address the characteristics of the drowsiness detection dataset. This model was designed with a balanced depth and layer configuration, tuned to detect drowsy versus awake states effectively. Its adaptability makes it highly suitable for this dataset, particularly given the unique combination of lighting conditions, facial obstructions, and varied camera angles. While it lacks the extensive prior knowledge embedded in pre-trained networks, this custom model benefits from being highly specialized and directly focused on the task requirements.

Pre-Trained Models vs. Custom CNN:

The use of pre-trained models such as ResNet50, InceptionV3, and VGG16 provides the advantage of leveraging large-scale learned features, which are helpful for generalization and reducing the training time required for feature learning. However, they may require more extensive fine-tuning to adapt effectively to specific attributes of this dataset, such as lighting variability and facial obstructions. In contrast, the custom CNN model, while lacking this large-scale pre-training, was designed specifically for this dataset, allowing it to be tuned more effectively to the task's unique demands. This tailored approach improves model efficiency and reduces training time, balancing complexity with computational feasibility.

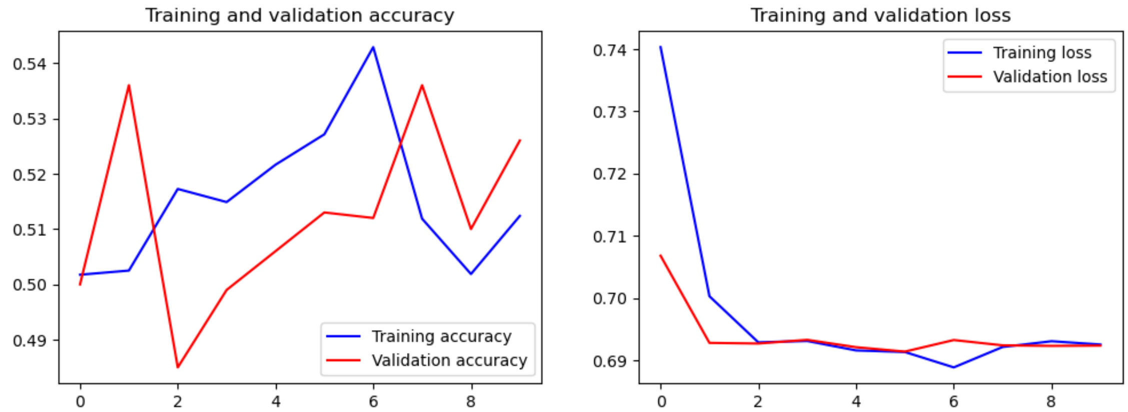
In conclusion, the combination of pre-trained and custom models provided a comprehensive basis for comparison, allowing for an in-depth understanding of the advantages and limitations of each model. This analysis informed decisions on the final model architecture, helping to ensure effective performance across diverse real-world conditions for driver drowsiness detection.

4.3 Comparative Analysis

This section presents a comparative analysis of the **Custom CNN model** and the **pre-trained models** (ResNet50, InceptionV3, and VGG16) used in the driver drowsiness detection task.

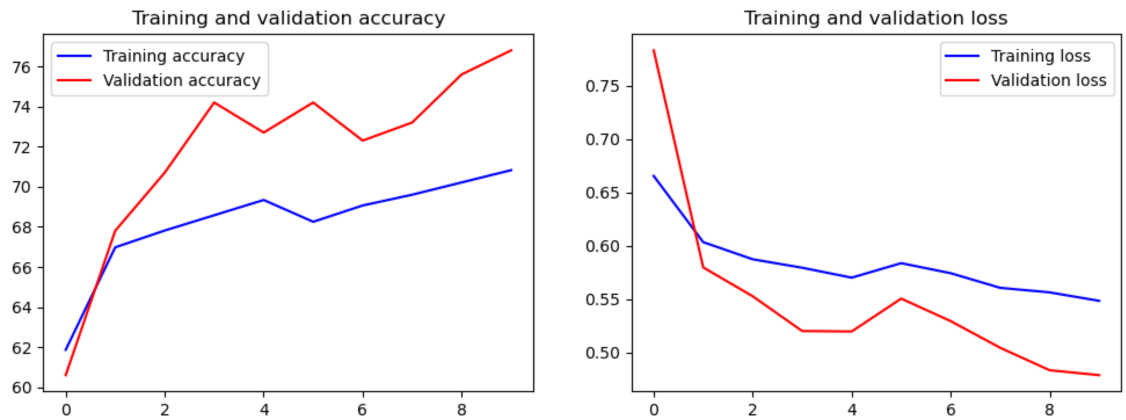
Graph for Resnet50:

Figure 3: Resnet50 Accuracies and Losses



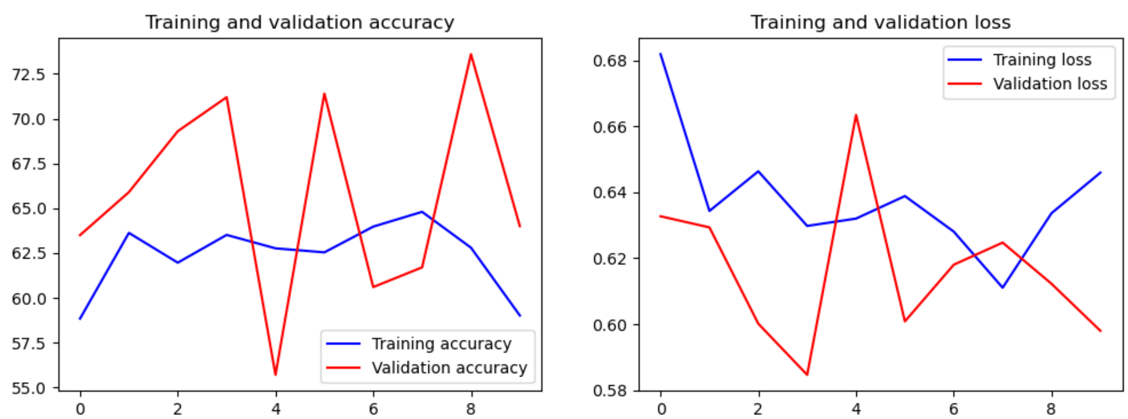
Graph for InceptionV3:

Figure 4: InceptionV3 Accuracies and Losses



Graph for VGG16:

Figure 5: VGG16 Accuracies and Losses



The models were evaluated on testing accuracy, testing loss, and key performance metrics for the critical “Drowsy” class, which is essential for assessing driver safety.

Table 1: Comparison of Test Values for all models

Model	Testing Accuracy	Testing Loss
ResNet50	55.60%	69.05%
InceptionV3	77.00%	48.11%
VGG16	62.60%	60.37%
Custom CNN	81.73%	31.72%

The **Custom CNN model** achieved the highest testing accuracy (81.73%) and the lowest testing loss (31.72%) among all models, demonstrating its superior performance in generalizing to new data. This is particularly notable given the high variation in lighting, facial obstructions, and camera angles in the dataset, suggesting that the custom CNN was more effective at capturing relevant features for drowsiness detection than the pre-trained architectures.

Table 2: Precision, Recall, and F1-Score Comparison (Drowsy Class)

Model	Precision (Drowsy)	Recall (Drowsy)	F1-Score (Drowsy)
ResNet50	0.54	0.82	0.65
InceptionV3	0.78	0.75	0.76
VGG16	0.93	0.27	0.42
Custom CNN	0.78	0.89	0.83

The **Custom CNN model** also excelled in terms of precision, recall, and F1-score for the "Drowsy" class. Specifically:

- **Precision**: Both InceptionV3 and Custom CNN maintained high precision scores, reflecting their effectiveness in accurately identifying drowsy drivers with minimal false positives.
- **Recall**: The Custom CNN model achieved the highest recall of 0.89, indicating its strength in detecting the maximum number of actual drowsy instances, which is essential for real-time driver monitoring systems.
- **F1-Score**: The Custom CNN's F1-score of 0.83, the highest among all models, suggests a balanced performance between precision and recall, affirming its reliability in detecting drowsy states effectively.

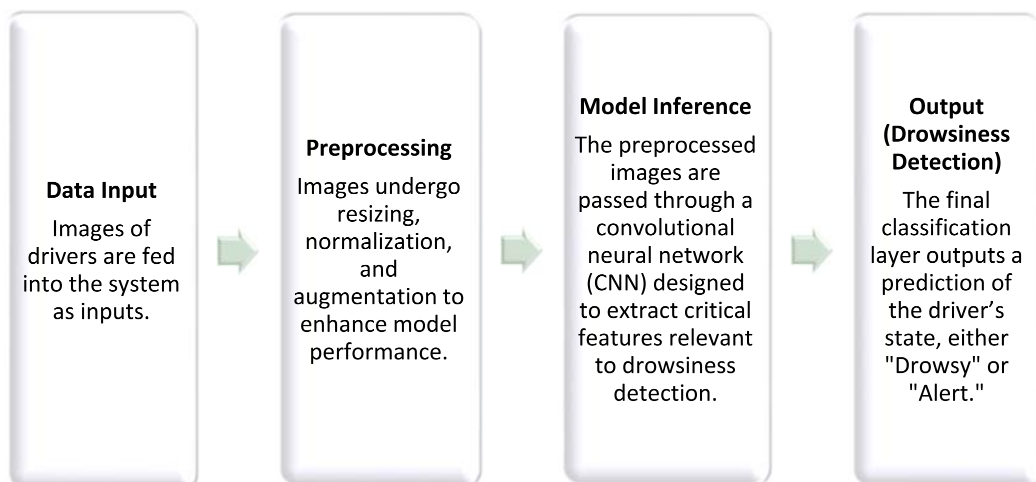
5. PROPOSED WORK

5.1 Solution Design

The driver drowsiness detection system was designed to accurately detect drowsy states in drivers based on facial image inputs. This section outlines the overall system architecture, key components, and the rationale for choosing a custom CNN over pre-trained models.

5.1.1 System Architecture

The system architecture follows a straightforward pipeline for image processing and classification.



5.1.2 Key Components

- **Preprocessing**:
  - Images were resized to a standard resolution suitable for the CNN input layer.
  - Normalization and augmentation (rotation, flipping) were applied to diversify the dataset, improving generalization.
- **Feature Extraction**:
  - Convolutional layers with ReLU activation capture key facial features and expressions indicative of drowsiness.
- **Classifier Design**:
  - The model's final layers included a fully connected network with softmax activation for two-class classification (Drowsy vs. Alert).

- **Additional Components**:
  - **Dropout** was used in multiple layers to prevent overfitting.
  - **Optimizer and Loss Function**: The Adam optimizer and categorical cross-entropy loss were chosen to optimize convergence while handling the imbalance of class distributions.

5.2 Implementation Details

This section details the technical implementation of the drowsiness detection system, including the programming environment, training techniques, and model training configuration.

5.2.1 Programming Languages and Libraries

1) **Languages**: Python
2) **Libraries**: TensorFlow, Keras (for model building and training), OpenCV (for image handling), and additional utility libraries like NumPy and Pandas for data management.
3) **Platforms**: Jupyter notebook, Google Colab

5.2.2 Data Preprocessing and Model Training

1) **Data Collection and Split**:

- The dataset comprises both drowsy and awake images, organized into two directories: Drowsy and Awake.
- Data is split into training, validation, and test sets at a ratio of 70:15:15:
  - Training set: 70% of the dataset
  - Validation set: 15% of the dataset
  - Test set: 15% of the dataset
- The split is achieved using train_test_split from sklearn, with stratification to ensure class balance across splits.

2) **Image Preprocessing and Augmentation**:

- Image Resolution and Colour Conversion:
  All images are resized to (224, 224) and converted to RGB format to standardize input dimensions.

- Face Detection and Cropping:

Face detection is performed using the Haar Cascade Classifier to focus on facial features, improving the model's relevance for drowsiness detection.

If faces are detected, the image is cropped to the face region; otherwise, the full image is used.

- Data Augmentation:
  Training images undergo random augmentations, including shear, zoom, horizontal flip, and brightness adjustments (range: 0.5–1.5), enhancing model generalization.

3) **Custom Data Generator**:

   A custom data generator class, FaceDetectionDataGenerator, iterates through images in batches, processes each image using face detection, and applies augmentation when enabled.

5.2.3 Architecture Design

1) **Model Design**:

   The CNN model is built sequentially with layers designed to extract increasingly abstract features, using the following configuration:

   Input shape: (224, 224, 3).

2) **Key elements of the architecture**:

- Convolutional Layers: Used for feature extraction with ReLU activation to introduce non-linearity.
- Max-Pooling Layers: Downsampled the feature maps, reducing computation and preventing overfitting.
- Fully Connected Layers: Mapped the learned features to the final output (Drowsy/Awake).
- Activation Functions: ReLU for hidden layers, and Softmax for the output layer to compute class probabilities.
- Optimizer: Adam optimizer was selected due to its adaptive learning rate and ability to handle sparse gradients.
- Loss Function: Categorical Cross-Entropy was chosen to calculate the difference between predicted probabilities and true labels.
- Dropout: Incorporated to mitigate overfitting, especially in fully connected layers.
- Epochs: The best results were obtained after 35 epochs, balancing accuracy and avoiding overfitting.
- Early Stopping: Monitors validation loss, with patience of 5 epochs, to avoid overfitting.

3) Model Summary:

Table 3: CNN Model Summary

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 222, 222, 32)	896
batch_normalization_4 (BatchNormalization)	(None, 222, 222, 32)	128
max_pooling2d_4 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_5 (Conv2D)	(None, 109, 109, 64)	18,496
batch_normalization_5 (BatchNormalization)	(None, 109, 109, 64)	256
max_pooling2d_5 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_6 (Conv2D)	(None, 52, 52, 128)	73,856
batch_normalization_6 (BatchNormalization)	(None, 52, 52, 128)	512
max_pooling2d_6 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_7 (Conv2D)	(None, 24, 24, 256)	295,168
batch_normalization_7 (BatchNormalization)	(None, 24, 24, 256)	1,024
max_pooling2d_7 (MaxPooling2D)	(None, 12, 12, 256)	0
flatten_1 (Flatten)	(None, 36,864)	0
dense_2 (Dense)	(None, 128)	4,718,720
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 1)	129

```
Total params: 5,109,187 (19.49 MB)
Trainable params: 5,108,225 (19.49 MB)
Non-trainable params: 960 (3.75 KB)
Optimizer params: 2 (12.00 B)
```

4) Epoch Variation Study

To optimize model performance, we experimented with various epoch settings. Training over multiple epochs revealed improvements in accuracy and stability, with peak performance at 35 epochs. At this point, the model balanced generalization without overfitting, demonstrating reliable results across the validation dataset.

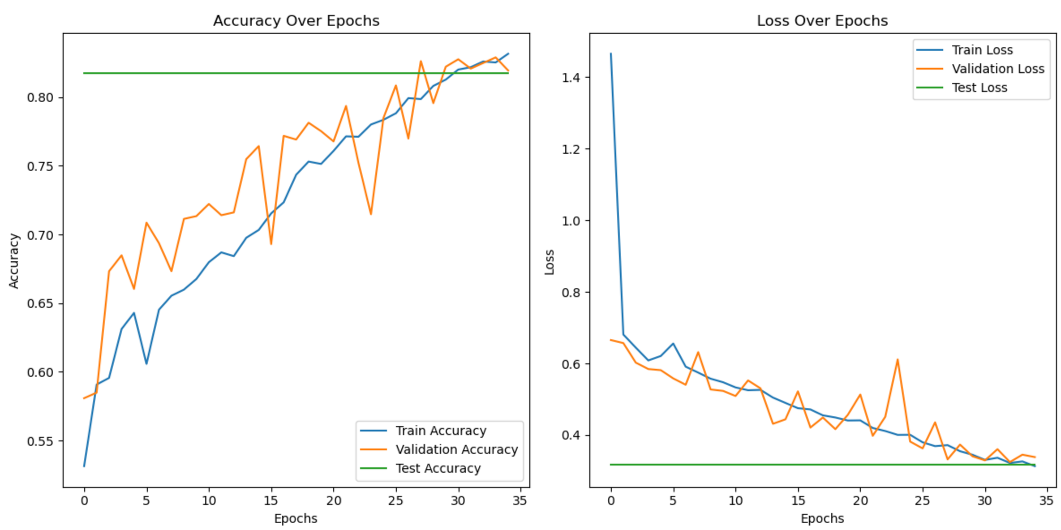
The model's performance improved steadily across the epochs. The table below summarizes the progression of accuracy and loss for both the training and validation sets over the course of 35 epochs:

Table 4: Training and Validation results over epochs

Epoch	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
1	0.5309	2.7755	0.5808	0.6653
5	0.6448	0.6106	0.6603	0.5813
10	0.6858	0.5405	0.7133	0.5233
15	0.6934	0.4951	0.7643	0.4440
20	0.7612	0.4372	0.7751	0.4564
25	0.7773	0.4050	0.7846	0.3818
30	0.8127	0.3448	0.8220	0.3407
35	0.8250	0.3109	0.8193	0.3385

From the table, it's clear that the model consistently improved in accuracy while the loss decreased, indicating successful learning. By the final epoch, the model achieved a training accuracy of 82.50% and a validation accuracy of 81.93%, with validation loss stabilizing at 0.3385.

Figure 6: Graph for accuracies and losses in CNN



5.3 Experiments and Results

Training Set Confusion Matrix:

- Precision: 0.90 (class 0), 0.81 (class 1)
- Recall: 0.79 (class 0), 0.91 (class 1)
- F1-score: 0.84 (class 0), 0.86 (class 1)
- Overall accuracy: **85%**

Validation Set Confusion Matrix:

- Precision: 0.87 (class 0), 0.79 (class 1)
- Recall: 0.77 (class 0), 0.88 (class 1)
- F1-score: 0.81 (class 0), 0.83 (class 1)
- Overall accuracy: **82%**

Test Set Confusion Matrix and Performance:

- Precision: 0.87 (class 0), 0.78 (class 1)
- Recall: 0.75 (class 0), 0.89 (class 1)
- F1-score: 0.80 (class 0), 0.83 (class 1)
- Overall accuracy: **81.73%**
- Loss: 0.3172

Table 5: Confusion Matrix for CNN Test dataset

	Awake (0)	Drowsy (1)
Awake (0)	554	185
Drowsy (1)	84	649

ROC Curve and AUC Score:

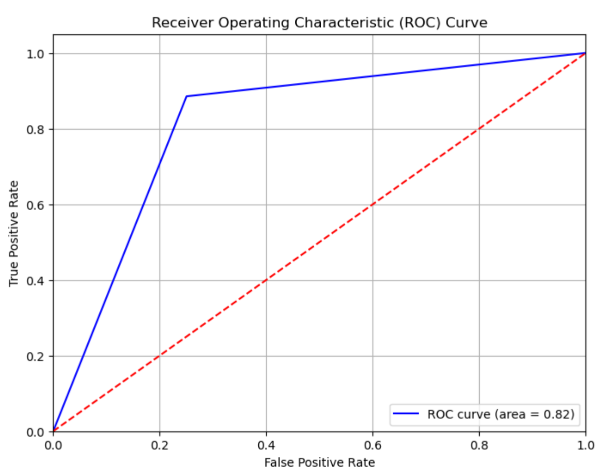


Figure 7: ROC and AUC for CNN Test dataset

The ROC curve demonstrates that the drowsiness detection model has a good overall performance, with an AUC of 0.82. This indicates that the model can effectively distinguish between drowsy and awake states while maintaining a balance between sensitivity and specificity.

Table 6: Final Comparision table for all models

Models	**ResNet50**	**InceptionV3**	**VGG16**	**Custom CNN**
Testing Accuracy	55.60%	77.00%	62.60%	81.73%
Validation Accuracy	52.60%	76.80%	64.00%	81.93%
Training Accuracy	51.24%	70.82%	59.01%	85%
Testing Loss	69.05%	48.11%	60.37%	31.72%
Validation Loss	69.24%	47.88%	59.80%	33.85%
Training Loss	69.25%	54.84%	64.60%	31.09%
Confusion Matrix	[[145, 355], [89, 411]]	[[393, 107], [123, 377]]	[[489, 11], [363, 137]]	[[554, 185], [84, 649]]
Precision (Awake)	0.62	0.76	0.57	0.87
Precision (Drowsy)	0.54	0.78	0.93	0.78
Recall (Awake)	0.29	0.79	0.98	0.75
Recall (Drowsy)	0.82	0.75	0.27	0.85
F1 Score (Awake)	0.40	0.77	0.72	0.80
F1 Score (Drowsy)	0.65	0.77	0.42	0.83

To conclude, the analysis highlights the custom CNN model as the top performer, achieving the highest testing accuracy (81.73%) with balanced precision and recall for both "Awake" and "Drowsy" classes. InceptionV3 follows closely, showing reliable accuracy and class distinction, particularly for "Awake." VGG16 performs well in "Awake" detection but struggles with "Drowsy" identification, while ResNet50's low accuracy and F1 scores indicate limitations in handling this dataset. Overall, the custom CNN, designed specifically for this task, proves optimal for driver drowsiness detection with reliable accuracy and robustness in diverse conditions.

6. CONCLUSION

This research project on Driver Drowsiness Detection aimed to develop a reliable model to distinguish between drowsy and alert drivers using a custom dataset and CNN model. The custom CNN achieved a commendable accuracy of 81.73% and an ROC AUC score of 0.82, demonstrating its ability to accurately identify features associated with drowsiness. This result underscores the effectiveness of the custom-designed convolutional architecture, which outperformed pre-trained models (ResNet50, VGG16, InceptionV3), specifically tailored for this application. Key architectural features, including convolutional layers with ReLU activation, max pooling, fully connected layers, the Adam optimizer, categorical cross-entropy loss, and dropout layers, allowed the model to capture subtle indications of drowsiness in static images effectively.

The dataset, a combination of 72.59% self-collected data (from 27 subjects, including 14 males, 13 females, and specific data of 300 images featuring a bus driver) and 27.41% external images, contributed significantly to the model's robustness. This mix provided a diverse and representative sample, enhancing the model's adaptability across various driving contexts and driver demographics. Despite these strengths, the model has certain limitations: it currently processes only static images, which restricts its immediate applicability to real-time scenarios like active driver monitoring. Furthermore, the model's slight overfitting suggests that an even more varied dataset is necessary for improved generalization across different driving environments.

Effectiveness of Custom CNN:

The custom CNN's tailored architecture and training approach were instrumental in achieving superior results. Its design specifically accommodated the variability in lighting, facial obstructions, and camera angles found in the dataset, allowing it to capture critical facial cues related to drowsiness more effectively than general pre-trained models. This adaptability proves the custom CNN to be particularly effective for detecting drowsiness from static images, setting a strong foundation for further advancements.

Current Limitations:

- The model currently processes only static images, limiting its use in real-time monitoring scenarios.
- Slight overfitting is observed, likely due to limited diversity within the dataset, affecting the model's generalizability.
- The model lacks the ability to process completely unseen data effectively, indicating a need for further tuning and dataset expansion.
- Real-time deployment could present challenges, as the current model setup may require optimization to handle continuous data feeds and ensure rapid response times.

Future Work:

- **Dataset Expansion:** Increase the dataset's demographic and environmental diversity to reduce overfitting and enhance the model's generalizability across different driving conditions.
- **Real-Time Video Analysis:** Develop the model to support real-time video processing, enabling dynamic detection and monitoring in active vehicle environments.
- **Advanced Techniques:** Explore sophisticated methods, such as transfer learning with video models or temporal data analysis, to further improve detection accuracy and responsiveness.
- **Optimization for Real-Time Use:** Optimize the model's architecture and processing capabilities to handle continuous data streams, ensuring fast response times essential for practical, on-road applications.

Final Remarks:

This project lays a promising foundation for enhancing driver safety through reliable drowsiness detection. While the current model successfully addresses static image recognition, advancing it to support real-time video processing would increase its practical utility and potentially prevent accidents due to drowsiness. This evolution would mark a significant step toward safer driving environments, contributing to safer roads for all.

REFERENCES

[1] Minhas, Abid Ali, Sohail Jabbar, Muhammad Farhan, and Muhammad Najam ul Islam. "A smart analysis of driver fatigue and drowsiness detection using convolutional neural networks." *Multimedia Tools and Applications* 81, no. 19 (2022): 26969-26986.

[2] Hashemi, Maryam, Alireza Mirrashid, and Aliasghar Beheshti Shirazi. "Driver safety development: Real-time driver drowsiness detection system based on convolutional neural network." *SN Computer Science* 1, no. 5 (2020): 289.

[3] Florez, Ruben, Facundo Palomino-Quispe, Roger Jesus Coaquira-Castillo, Julio Cesar Herrera-Levano, Thuanne Paixão, and Ana Beatriz Alvarez. "A cnn-based approach for driver drowsiness detection by real-time eye state identification." *Applied Sciences* 13, no. 13 (2023): 7849.

[4] Salman, Rais Mohammad, Mahbubur Rashid, Rupal Roy, Md Manjurul Ahsan, and Zahed Siddique. "Driver drowsiness detection using ensemble convolutional neural networks on YawDD." *arXiv preprint arXiv:2112.10298* (2021).

[5] Jebraeily, Yashar, Yousef Sharafi, and Mohammad Teshnehlab. "Driver Drowsiness Detection Based on Convolutional Neural Network Architecture Optimization Using Genetic Algorithm." *IEEE Access* (2024).

[6] Salem, Dina, and Mohamed Waleed. "Drowsiness detection in real-time via convolutional neural networks and transfer learning." *Journal of Engineering and Applied Science* 71, no. 1 (2024): 122.

[7] Altameem, Ayman, Ankit Kumar, Ramesh Chandra Poonia, Sandeep Kumar, and Abdul Khader Jilani Saudagar. "Early identification and detection of driver drowsiness by hybrid machine learning." *IEEE Access* 9 (2021): pp.162805-162819.

[8] Albadawi, Yaman, Maen Takruri, and Mohammed Awad. "A review of recent developments in driver drowsiness detection systems." *Sensors* 22, no. 5 (2022): 2069.

[9] Albadawi, Yaman, Aneesa AlRedhaei, and Maen Takruri. "Real-time machine learning-based driver drowsiness detection using visual features." *Journal of imaging* 9, no. 5 (2023): 91.

[10] Jabbar, Rateb, Mohammed Shinoy, Mohamed Kharbeche, Khalifa Al-Khalifa, Moez Krichen, and Kamel Barkaoui. "Driver drowsiness detection model using convolutional neural networks techniques for android application." In *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT)*, pp. 237-242. IEEE, 2020.

[11] Suvarna Nandyal, Sharanabasappa. "Original Research Article Deep ResNet 18 and enhanced firefly optimization algorithm for on-road vehicle driver drowsiness detection." *Journal of Autonomous Intelligence* 7, no. 3 (2024).

[12] Díaz-Santos, Sonia, Óscar Cigala-Álvarez, Ester Gonzalez-Sosa, Pino Caballero-Gil, and Cándido Caballero-Gil. "Driver identification and detection of drowsiness while driving." *Applied Sciences* 14, no. 6 (2024): 2603.

[13] Jahan, Israt, KM Aslam Uddin, Saydul Akbar Murad, M. Saef Ullah Miah, Tanvir Zaman Khan, Mehedi Masud, Sultan Aljahdali, and Anupam Kumar Bairagi. "4D: a real-time driver drowsiness detector using deep learning." *Electronics* 12, no. 1 (2023): 235.

[14] Ahmed, Mohammed Imran Basheer, Halah Alabdulkarem, Fatimah Alomair, Dana Aldossary, Manar Alahmari, Munira Alhumaidan, Shoog Alrassan, Atta Rahman, Mustafa Youldash, and Gohar Zaman. "A deep-learning approach to driver drowsiness detection." *Safety* 9, no. 3 (2023): 65.

[15] Mandal, Bappaditya, Liyuan Li, Gang Sam Wang, and Jie Lin. "Towards detection of bus driver fatigue based on robust visual analysis of eye state." *IEEE Transactions on Intelligent Transportation Systems* 18, no. 3 (2016): 545-557.

[16] Akrout, Belhassen, and Walid Mahdi. "A novel approach for driver fatigue detection based on visual characteristics analysis." Journal of Ambient Intelligence and Humanized Computing 14, no. 1 (2023): 527-552.

[17] Dipu, Md Tanvir Ahammed, Syeda Sumbul Hossain, Yeasir Arafat, and Fatama Binta Rafiq. "Real-time driver drowsiness detection using deep learning." *International Journal of Advanced Computer Science and Applications* 12, no. 7 (2021).

[18] Rajkar, Ajinkya, Nilima Kulkarni, and Aniket Raut. "Driver drowsiness detection using deep learning." In *Applied Information Processing Systems: Proceedings of ICCET 2021*, pp. 73-82. Springer Singapore, 2022.

[19] Pandey, Nageshwar Nath, and Naresh Babu Muppalaneni. "Real-time drowsiness identification based on eye state analysis." In *2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)*, pp. 1182-1187. IEEE, 2021.

[20] Jabbar, Rateb, Khalifa Al-Khalifa, Mohamed Kharbeche, Wael Alhajyaseen, Mohsen Jafari, and Shan Jiang. "Real-time driver drowsiness detection for android application using deep neural networks techniques." Procedia computer science 130 (2018): 400-407.

[21] Yılmaz, Esra Kavalcı, and M. Ali Akcayol. "SUST-DDD: A real-drive dataset for driver drowsiness detection." In Conference of Open Innovations Association (FRUCT), vol. 6. 2022.

[22] Dipu, Md Tanvir Ahammed, Syeda Sumbul Hossain, Yeasir Arafat, and Fatama Binta Rafiq. "Real-time driver drowsiness detection using deep learning." *International Journal of Advanced Computer Science and Applications* 12, no. 7 (2021).

[23] Chirra, Venkata Rami Reddy, Srinivasulu Reddy Uyyala, and Venkata Krishna Kishore Kolli. "Deep CNN: A Machine Learning Approach for Driver Drowsiness Detection Based on Eye State." *Rev. d'Intelligence Artif.* 33, no. 6 (2019): 461-466.

[24] Jabbar, Rateb, Mohammed Shinoy, Mohamed Kharbeche, Khalifa Al-Khalifa, Moez Krichen, and Kamel Barkaoui. "Driver drowsiness detection model using convolutional neural networks techniques for android application." In *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT)*, pp. 237-242. IEEE, 2020.

[25] Salman, Rais Mohammad, Mahbubur Rashid, Rupal Roy, Md Manjurul Ahsan, and Zahed Siddique. "Driver drowsiness detection using ensemble convolutional neural networks on YawDD." *arXiv preprint arXiv:2112.10298* (2021).

[26] https://www.kaggle.com/datasets/ismailnasri20/driver-drowsiness-dataset-ddd

[27] https://www.kaggle.com/datasets/nehalmahajan28/driver-drowsiness-detection-dataset

DRIVER DROWSINESS DETECTION

Lavi Garg
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
21ituon138@ddu.ac.in

Charmi Parmar
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
21itubs073@ddu.ac.in

Prof. Archana N. Vyas
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
anvyas.it@ddu.ac.in

***Abstract*—Driver drowsiness is a major contributor to traffic accidents, highlighting the need for effective detection systems to improve road safety. This research presents a Driver Drowsiness Detection model based on a custom Convolutional Neural Network (CNN) architecture, designed to identify drowsiness from static facial images. The dataset used for training and evaluation was a combination of self-collected images from real-world conditions, including bus drivers, and publicly available datasets to ensure diversity. The custom CNN model was developed to achieve high accuracy in detecting drowsiness, reaching an accuracy of 81.73%. The implementation of this model can help reduce the risk of accidents and improve overall road safety.**

Keywords—Convolutional Neural Network (CNN), Road Safety, Deep Learning, Image Classification, Accident Prevention, Custom CNN Model

I. INTRODUCTION

Driver drowsiness is a significant contributor to road accidents worldwide, making its early detection crucial for enhancing road safety. When drivers experience drowsiness, their attention and reaction times are severely impaired, increasing the likelihood of accidents. Traditional methods of detecting driver drowsiness, such as manual observation or physiological sensors, are often limited by human error, invasiveness, and the difficulty of monitoring drivers in real-time. To overcome these challenges, this research focuses on leveraging deep learning and computer vision technologies to develop a non-invasive and accurate system for detecting driver drowsiness through facial images. This approach offers a practical and scalable solution to address a widespread issue, potentially reducing accidents and saving lives.

The proposed system employs Convolutional Neural Networks (CNNs) to analyze facial features and detect signs of drowsiness. CNNs are highly effective for image classification tasks due to their ability to automatically extract relevant features from raw image data. In this research, facial cues such as eye closure, yawning, and head positioning are used to distinguish between drowsy and awake drivers. The system's ability to work with static images makes it an efficient solution for continuous monitoring without the need for specialized sensors or active driver involvement. This allows for seamless integration into existing in-vehicle systems, ensuring minimal disruption to the driver while maintaining reliable performance in real-time.

The system is trained using a custom dataset, which includes images collected from bus drivers under real-world conditions, such as varying lighting, facial occlusions, and demographic diversity. Additionally, publicly available datasets are incorporated to enhance the model's ability to generalize to different driving environments. The use of facial features enables the model to reliably classify drivers as either drowsy or awake, making it a suitable tool for early detection and intervention in real-time situations. This comprehensive dataset helps the model adapt to various challenges encountered in practical applications, ensuring robustness and accuracy across different driving scenarios.

The research also focuses on improving the robustness of the model to handle different environmental factors that can affect the accuracy of drowsiness detection. Factors such as lighting changes, face orientations, and the presence of facial accessories are considered to ensure that the system performs well across diverse conditions. By accounting for these challenges, the system is designed to be reliable even in complex, real-world driving environments, enhancing its potential for widespread use. This level of adaptability is crucial for ensuring the system's effectiveness in a variety of settings, making it a versatile solution for both commercial and private vehicles.

The primary goal of this research is to develop a system that is both accurate and non-intrusive, capable of detecting driver drowsiness with high reliability. The scope of the project involves data collection, model development, and thorough evaluation using performance metrics such as accuracy, precision, recall, and F1-score. By incorporating this system into vehicles, it has the potential to reduce the number of accidents caused by drowsiness, ultimately contributing to improved road safety and helping prevent drowsy driving-related incidents. With further development, this system could become an essential safety feature in modern vehicles, significantly enhancing driver safety.

II. LITERATURE REVIEW

First Driver drowsiness detection has been an essential research area due to its critical role in enhancing road safety. Various methods and systems have been proposed over the years to detect and prevent accidents caused by driver drowsiness. Among the most commonly used techniques are those based on facial expressions, eye movements, head poses, and other indicators.

Ayman Altameem et al. [1] used Support Vector Machines (SVM) to detect facial expression changes, achieving 83.25% accuracy under varying lighting conditions. This demonstrated the potential of facial expressions for real-time detection.

Yaman Albadawi et al. [2] combined Eye Aspect Ratio (EAR) and head pose estimation, achieving 99% accuracy with classifiers like random forest and SVM. This approach enhanced the reliability of real-time drowsiness detection.

Hashemi et al. [3] introduced a CNN-based system focusing on eye closure detection. Their model, tested on the ZJU dataset, achieved 98.15% accuracy and low computational complexity, making it suitable for real-time applications.

Rateb Jabbar et al. [4] developed a CNN-based driver drowsiness detection system using facial landmark detection. Their model achieved 83.3% accuracy, utilizing the NTHU Driver Drowsiness Detection dataset. Designed for mobile

979-8-3503-4846-0/24/$31.00 ©2024 IEEE

and embedded systems, it demonstrated real-time feasibility and enhanced road safety.

Salman et al. [5] proposed an ensemble CNN (ECNN) model that incorporated facial features like yawning and eye closure. Their system achieved 99.42% accuracy on the YawDD dataset, proving that ensemble learning can improve robustness in drowsiness detection.

Md. Tanvir Ahammed Dipu et al. [6] used MobileNet with Single Shot Multibox Detector (SSD) for detecting yawning and eye states, achieving a mean average precision (mAP) of 98.02%. This lightweight model was ideal for real-time detection.

Minhas et al. [7] tested InceptionV3, VGG16, and ResNet50 on a custom dataset, with ResNet50 achieving the highest accuracy of 93.69%. Their research demonstrated how CNNs can be adapted for diverse environmental conditions.

Jahan et al. [8] created a driver drowsiness detection system using a custom 4D CNN model, along with VGG16 and VGG19. The 4D model achieved 97.53% accuracy, while the VGG models reached 95.93% and 95.03%, respectively.

Esra Kavalcı Yılmaz and M. Ali Akcayol [9] introduced a hybrid VGG19+LSTM model, achieving 90.53% accuracy on the SUST-DDD dataset. This approach underscored the role of hybrid models in enhancing drowsiness detection.

Florez et al. [10] utilized ResNet50V2 for drowsiness detection and achieved 99.71% accuracy on the NITYMED dataset. Their use of transfer learning and MediaPipe for eye detection improved real-time performance.

Mohammed Imran Basheer Ahmed et al. [11] focused on facial expression analysis using CNNs and VGG16. Their system, tested on 2,900 images, achieved 97% accuracy, demonstrating CNN's potential in real-time detection.

Salem and Waleed [12] used MobileNetV2 and InceptionV3 models to develop a highly effective system, achieving accuracies of 99.94% and 99.82% on the Driver Drowsiness Dataset, proving their real-time applicability.

III. Proposed Work

A. Dataset

The dataset used for this project is a combination of a custom-created dataset and publicly available sources from Kaggle. By merging these sources, the dataset gained substantial diversity, making it more applicable to real-world scenarios. It includes images captured under different lighting conditions (day and night), with facial obstructions like sunglasses, specs, masks, and hats, and from various camera angles, simulating a typical driving environment. The data is evenly distributed between two classes—Awake and Drowsy, with 5,000 images in each category.

A total of 7,258 images were self-collected from 27 subjects (14 male, 13 female) aged between 20 and 60 years, with some images specifically of bus drivers (300 images) to simulate different vehicle types. The remaining 2,742 images were sourced externally to complement the custom data. The combination of self-collected and external data ensures a balanced and varied dataset, allowing the model to generalize effectively in diverse conditions. This diverse dataset aims to improve the model's ability to detect drowsiness accurately across a wide range of real-world driving environments.use abbreviations in the title or heads unless they are unavoidable.

B. Model Analysis

In our driver drowsiness detection system, we experimented with both pre-trained models (ResNet50, InceptionV3, and VGG16) and a custom-designed Convolutional Neural Network (CNN) tailored specifically for this task. The pre-trained models were fine-tuned on our dataset, leveraging their extensive feature extraction capabilities, which provided a strong baseline. These models benefited from their ability to capture complex patterns due to their deep architectures, having been trained on vast image datasets like ImageNet. However, while the pre-trained models showed promising results in terms of accuracy, their performance was constrained by the fact that they were not initially designed for detecting driver drowsiness. The transfer learning approach provided a quick way to adapt these models to the task, but the generalized features they learned did not fully capture the specific nuances of the drowsiness detection problem.

However, the custom CNN, which was specifically built for this project, outperformed the pre-trained models in terms of accuracy and classification performance. This custom architecture was optimized to capture the unique facial features associated with drowsiness, such as eye closure patterns, yawning, and subtle head movements. By fine-tuning the architecture and hyperparameters, the custom CNN was able to learn task-specific features more effectively, leading to more accurate predictions of 'Awake' and 'Drowsy' states. Additionally, the custom model was able to adapt better to the diverse environmental conditions present in the dataset, such as varying lighting and facial orientations, which made it more robust in real-world scenarios.

C. System Architecture

The system architecture for the driver drowsiness detection system is designed as a streamlined pipeline that processes input images to determine the driver's alertness. It starts with the data input phase, where facial images of the driver are captured in real-time through an onboard camera. These images are then passed through a preprocessing stage, where they are resized, normalized, and adjusted for variations such as lighting conditions and facial orientations to ensure consistency. The preprocessed images are then fed into the model for inference, where a Convolutional Neural Network (CNN), either custom-built or pre-trained, analyzes the facial features—such as eye closure and head position—to detect signs of drowsiness. It comprises three key components: data input, preprocessing, and model inference, followed by the final output, where the system classifies the driver as either 'drowsy' or 'awake.'

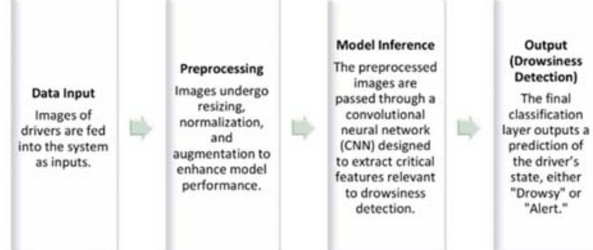


Figure 1 - System Architecture

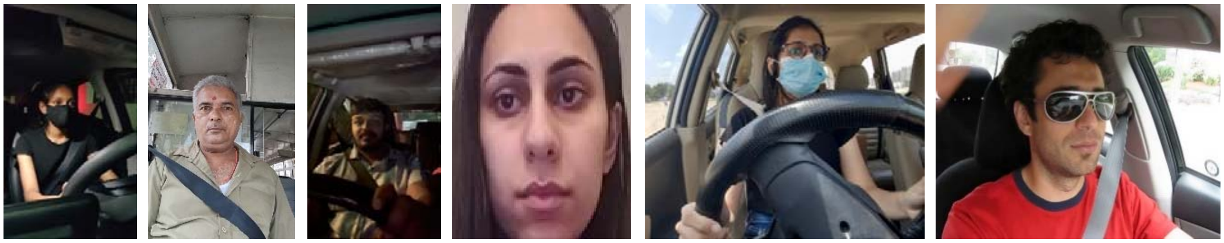


Figure 2 - Awake

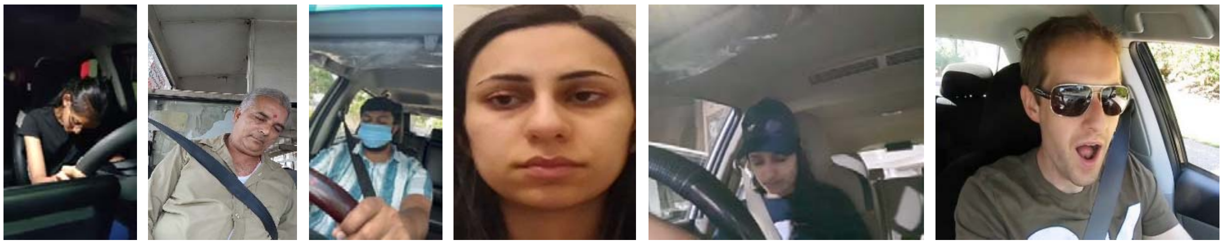


Figure 3 - Drowsy

D. Data Preprocessing and Model Training

a) Data Collection and Split

The dataset contains images of drivers in two states: Drowsy and Awake, and is divided into three sets: 70% for training, 15% for validation, and 15% for testing. The data is split using the train_test_split function from scikit-learn.

b) Image Preprocessing and Augmentation

All images are resized to 224x224 pixels and converted to RGB format for consistency. Face detection is performed using the Haar Cascade Classifier to focus on facial features, and if a face is detected, the image is cropped to highlight the face region; otherwise, the full image is used. To enhance the model's generalization capabilities, data augmentation techniques such as shear, zoom, horizontal flip, and brightness adjustments (ranging from 0.5 to 1.5) are applied to the training set, allowing the model to better handle variations in real-world scenarios.

E. Architecture Design

The CNN model for driver drowsiness detection follows a sequential architecture, designed to progressively extract and process features from input images. The model takes images of size (224, 224, 3) as input and applies a series of convolutional layers to learn and capture local features such as edges, textures, and shapes. These layers are followed by ReLU (Rectified Linear Unit) activation functions, which introduce non-linearity into the model, allowing it to model more complex relationships between the pixels in the image. To manage the size of the data and reduce computational load, max-pooling layers are employed after each convolution. These layers downsample the feature maps by selecting the maximum values in a given window, which also helps to minimize overfitting by preserving only the most important features.

As the data moves deeper into the model, the fully connected layers take the extracted features and map them to the final prediction. The model uses Softmax in the output layer to assign probabilities to each class—"Drowsy" or "Awake." To further optimize training, the Adam optimizer is chosen for its adaptive learning rate capabilities and its efficiency in dealing with sparse gradients. Categorical cross-entropy is used as the loss function, comparing the predicted output probabilities with the true labels to measure the model's performance. This choice of loss function is well-suited for multi-class classification problems like this one.

To prevent overfitting, especially in the fully connected layers, dropout layers are included, randomly disabling a fraction of neurons during training to encourage generalization. The model was trained for 35 epochs, which provided a balance between achieving high accuracy and avoiding overfitting. Early stopping was also implemented, monitoring validation loss with a patience of 5 epochs. This mechanism ensures that the model halts training once performance starts to degrade, ensuring the best possible generalization to unseen data without excessive training.

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 222, 222, 32)	896
batch_normalization_4 (BatchNormalization)	(None, 222, 222, 32)	128
max_pooling2d_4 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_5 (Conv2D)	(None, 109, 109, 64)	18,496
batch_normalization_5 (BatchNormalization)	(None, 109, 109, 64)	256
max_pooling2d_5 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_6 (Conv2D)	(None, 52, 52, 128)	73,856
batch_normalization_6 (BatchNormalization)	(None, 52, 52, 128)	512
max_pooling2d_6 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_7 (Conv2D)	(None, 24, 24, 256)	295,168
batch_normalization_7 (BatchNormalization)	(None, 24, 24, 256)	1,024
max_pooling2d_7 (MaxPooling2D)	(None, 12, 12, 256)	0
flatten_1 (Flatten)	(None, 36,864)	0
dense_2 (Dense)	(None, 128)	4,718,720
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 1)	129

Total params: 5,109,187 (19.49 MB)
Trainable params: 5,108,225 (19.49 MB)
Non-trainable params: 960 (3.75 KB)
Optimizer params: 2 (12.00 B)

Figure 4 - CNN Architecture

IV. Experimental Results

The experiments conducted to evaluate the performance of the driver drowsiness detection model were performed using the custom CNN model, as well as several pre-trained models, including ResNet50, InceptionV3, and VGG16. The evaluation metrics used for assessing model performance include precision, recall, F1-score, accuracy, and loss for the training, validation, and test datasets.

The model's performance evolved over 35 epochs, as shown in the following analysis of training and validation accuracy and loss. Training accuracy steadily increased, reaching 82.50% by the 35th epoch, while training loss decreased and stabilized at 0.8250, indicating effective learning without overfitting. Validation accuracy followed a similar trend, ending at 81.93%, with validation loss gradually decreasing to 0.3385, suggesting strong generalization to unseen data. By the 35th epoch, the model had achieved a good balance between accuracy and loss, making it suitable for deployment in detecting driver drowsiness.

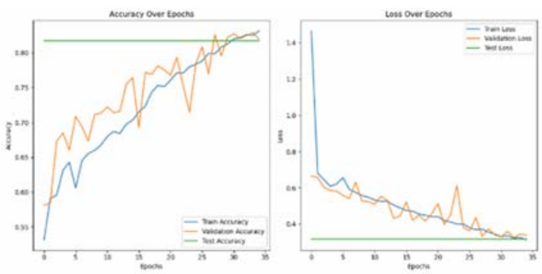


Figure 5 - Graph for accuracies and losses in CNN

The custom CNN model showed strong performance across training, validation, and test sets. During training, it effectively identified both "Awake" and "Drowsy" states with balanced precision and recall, suggesting good generalization to the training data. The model achieved solid accuracy and consistent F1-scores, indicating effective learning.

In the validation phase, the model maintained balanced precision and recall, excelling in detecting "Drowsy" states. The validation accuracy confirmed the model’s ability to generalize well to unseen data, with slightly lower precision for "Drowsy," but overall consistent results.

On the test set, the model performed well, with high accuracy and a good balance between precision and recall. It showed greater sensitivity to identifying "Drowsy" states, although precision for "Drowsy" was lower than for "Awake." Overall, the model demonstrated robust generalization, making it suitable for real-world applications.

The confusion matrix for the test set reflects the model's ability to differentiate between "Awake" and "Drowsy" states. It shows the model's precision and recall performance for each class, indicating its ability to correctly identify both "Awake" and "Drowsy" drivers. The overall accuracy of the model on the test set is 81.73%, with a loss of 0.3172, highlighting its solid performance in real-world data classification.

	Awake (0)	Drowsy (1)
Awake (0)	554	185
Drowsy (1)	84	649

Table 1 - Confusion Matrix for CNN Test dataset

The ROC curve for the test dataset demonstrates the model's ability to distinguish between "Awake" and "Drowsy" states. The curve shows the trade-off between sensitivity (True Positive Rate) and specificity (1 - False Positive Rate). A higher AUC (Area Under the Curve) indicates better performance. For this model, the ROC curve reveals an AUC of 0.82, suggesting that the CNN model effectively distinguishes between the two states while maintaining a good balance between false positives and false negatives. This confirms the model's robustness and its ability to generalize well on unseen test data.

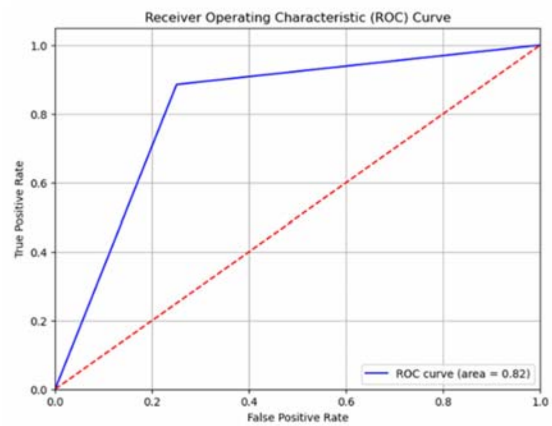


Figure 6 - ROC and AUC for CNN Test dataset

In addition to evaluating the custom CNN model, I also analyzed the performance of pre-trained models, including ResNet50, VGG16, and InceptionV3. These models were fine-tuned for the driver drowsiness detection task, and their performance was compared with the custom CNN. Each model's precision, recall, F1-score, and overall accuracy were recorded to understand their strengths and limitations.

Models	ResNet50	InceptionV3	VGG16	Custom CNN
Testing Accuracy	55.60%	77.00%	62.60%	81.73%
Validation Accuracy	52.60%	76.80%	64.00%	81.93%
Training Accuracy	51.24%	70.82%	59.01%	85%
Testing Loss	69.05%	48.11%	60.37%	31.72%
Validation Loss	69.24%	47.88%	59.80%	33.85%
Training Loss	69.25%	54.84%	64.60%	31.09%
Confusion Matrix	[[145, 355], [89, 411]]	[[393, 107], [123, 377]]	[[489, 11], [363, 137]]	[[554, 185], [84, 649]]
Precision (Awake)	0.62	0.76	0.57	0.87
Precision (Drowsy)	0.54	0.78	0.93	0.78
Recall (Awake)	0.29	0.79	0.98	0.75
Recall (Drowsy)	0.82	0.75	0.27	0.85
F1 Score (Awake)	0.40	0.77	0.72	0.80
F1 Score (Drowsy)	0.65	0.77	0.42	0.83

Table 2 - Final Comparision table for all models

The confusion matrices for all models highlight these performance differences. For example, ResNet50 and VGG16 had more misclassifications for "Drowsy" states, while InceptionV3 performed better but still fell short compared to the custom CNN. The custom CNN showed the most balanced confusion matrix, with fewer misclassifications in both "Awake" and "Drowsy" categories, making it the most reliable model for this task. These findings suggest that while pre-trained models provide a good starting point, the custom CNN achieves the best results.

To conclude, the analysis demonstrates that the custom CNN model outperforms others, achieving the highest testing accuracy of 81.73% with balanced precision and recall for both "Awake" and "Drowsy" classes. InceptionV3 also performs well, showing reliable accuracy and particularly excelling in detecting "Awake" states. VGG16 achieves good results for "Awake" detection but faces challenges with accurately identifying "Drowsy" states. ResNet50, however, shows lower accuracy and F1 scores, indicating limitations when applied to this dataset. Overall, the custom CNN model, tailored specifically for driver drowsiness detection, proves to be the most optimal, delivering reliable accuracy and robustness across various conditions.

V. Conclusion

This research project on Driver Drowsiness Detection successfully developed a custom CNN model that achieved an accuracy of 81.73% and an ROC AUC score of 0.82, showcasing its ability to identify drowsy drivers effectively. The custom CNN outperformed pre-trained models like ResNet50, VGG16, and InceptionV3, thanks to its tailored design. Key features such as convolutional layers with ReLU activation, max pooling, and dropout layers enabled the model to capture subtle facial cues related to drowsiness. The dataset, which included 72.59% self-collected data and 27.41% external images, ensured diversity and robustness. However, the model is limited to static image processing, which restricts its immediate use in real-time monitoring. Additionally, slight overfitting indicates the need for a more varied dataset to enhance generalization across different driving environments.

Future work will focus on expanding the dataset's diversity to improve generalization and reduce overfitting. The transition from static image processing to real-time video analysis is a key goal, enabling dynamic detection in active vehicle environments. Advanced techniques, such as transfer learning with video models and temporal data analysis, will be explored to improve accuracy and responsiveness. Additionally, optimizing the model's architecture for real-time deployment will ensure it can handle continuous data streams efficiently. By achieving these goals, the project has the potential to significantly enhance driver safety and prevent accidents caused by drowsiness. The next steps will ensure the model's practical application in real-world driving conditions, contributing to safer roads for all.

VI. References

[1] Altameem, Ayman, Ankit Kumar, Ramesh Chandra Poonia, Sandeep Kumar, and Abdul Khader Jilani Saudagar. "Early identification and detection of driver drowsiness by hybrid machine learning." IEEE Access 9 (2021): pp.162805-162819.

[2] Albadawi, Yaman, Aneesa AlRedhaei, and Maen Takruri. "Real-time machine learning-based driver drowsiness detection using visual features." Journal of imaging 9, no. 5 (2023): 91.

[3] Hashemi, Maryam, Alireza Mirrashid, and Aliasghar Beheshti Shirazi. "Driver safety development: Real-time driver drowsiness detection system based on convolutional neural network." SN Computer Science 1, no. 5 (2020): 289.

[4] Pandey, Nageshwar Nath, and Naresh Babu Muppalaneni. "Real-time drowsiness identification based on eye state analysis." In 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), pp. 1182-1187. IEEE, 2021.

[5] Salman, Rais Mohammad, Mahbubur Rashid, Rupal Roy, Md Manjurul Ahsan, and Zahed Siddique. "Driver drowsiness detection using ensemble convolutional neural networks on YawDD." arXiv preprint arXiv:2112.10298 (2021).

[6] Dipu, Md Tanvir Ahammed, Syeda Sumbul Hossain, Yeasir Arafat, and Fatama Binta Rafiq. "Real-time driver drowsiness detection using deep learning." International Journal of Advanced Computer Science and Applications 12, no. 7 (2021).

[7] Minhas, Abid Ali, Sohail Jabbar, Muhammad Farhan, and Muhammad Najam ul Islam. "A smart analysis of driver fatigue and drowsiness detection using convolutional neural networks." Multimedia Tools and Applications 81, no. 19 (2022): 26969-26986.

[8] Jahan, Israt, KM Aslam Uddin, Saydul Akbar Murad, M. Saef Ullah Miah, Tanvir Zaman Khan, Mehedi Masud, Sultan Aljahdali, and Anupam Kumar Bairagi. "4D: a real-time driver drowsiness detector using deep learning." Electronics 12, no. 1 (2023): 235.

[9] Yılmaz, Esra Kavalcı, and M. Ali Akcayol. "SUST-DDD: A real-drive dataset for driver drowsiness detection." In Conference of Open Innovations Association (FRUCT), vol. 6. 2022.

[10] Florez, Ruben, Facundo Palomino-Quispe, Roger Jesus Coaquira-Castillo, Julio Cesar Herrera-Levano, Thuanne Paixão, and Ana Beatriz Alvarez. "A cnn-based approach for driver drowsiness detection by real-time eye state identification." Applied Sciences 13, no. 13 (2023): 7849.

[11] Ahmed, Mohammed Imran Basheer, Halah Alabdulkarem, Fatimah Alomair, Dana Aldossary, Manar Alahmari, Munira Alhumaidan, Shoog Alrassan, Atta Rahman, Mustafa Youldash, and Gohar Zaman. "A deep-learning approach to driver drowsiness detection." Safety 9, no. 3 (2023): 65.

[12] Salem, Dina, and Mohamed Waleed. "Drowsiness detection in real-time via convolutional neural networks and transfer learning." Journal of Engineering and Applied Science 71, no. 1 (2024): 122.

CURRICULUM VITAE

Academic Details

Graduation: Bachelor of Technology

Branch: Information Technology

University: Dharmsinh Desai University

Place: Nadiad

Personal Details

Name: Lavi Garg

Address: Memnagar, Ahmedabad

Contact No.: 6355617251

Email: lavi.garg004@gmail.com

Name: Charmi Parmar

Address: Nikol, Ahmedabad

Contact No.: 8140557110

Email: ccparmar2004@gmail.com